

Two Surveys, Two Biases

Topic 3.4 · TODAY'S PROBLEM

Suppose a town studies support for a park fee.

Mail survey: A neutral question is mailed to 500 randomly chosen households. Only 120 return it; 78% of respondents oppose the fee.

Phone survey: The town calls 300 random phone numbers. Of 240 people who answer, 62% oppose after hearing, *Do you support a NEW FEE that would tax park users?*

One council member says 62% must be closer to the truth because more people answered. Another says 78% must be closer because the households were randomly chosen.

Park-fee surveys

	<i>n</i>	Reply	Opp.	Flaw
Mail	500	120	78%	Nonresp.
Phone	300	240	62%	Wording

FIRST TAKE (5 min, on your sheet)

Which estimate, if either, would you report to the town? Name one detail behind your choice.

- DOK 1** (a) Compute the response rate for each survey. Name the main source of bias in each.
- DOK 2** (b) Explain how each bias could change the reported percent opposing the fee. State a likely direction when the information supports one.
- DOK 3** (c) Decide whether either council member has established which estimate is closer to the town's true opinion. Cite the 24% response rate and the loaded wording in your reasoning, explain why neither estimate is trustworthy as is, and propose one design-specific fix for each survey.

Video + follow-along: u3_lesson4_live.html (scan the code)

Finish (a)–(c) after the video. Turn in your sheet.

